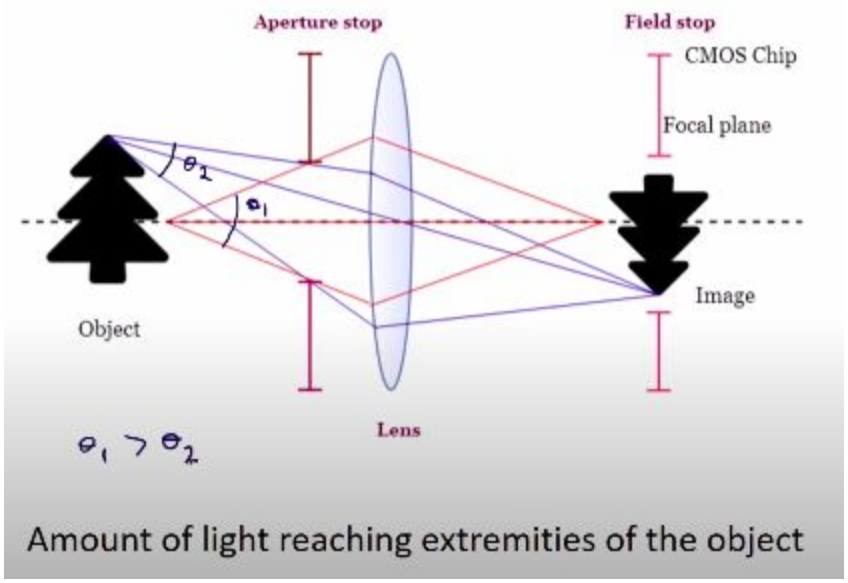
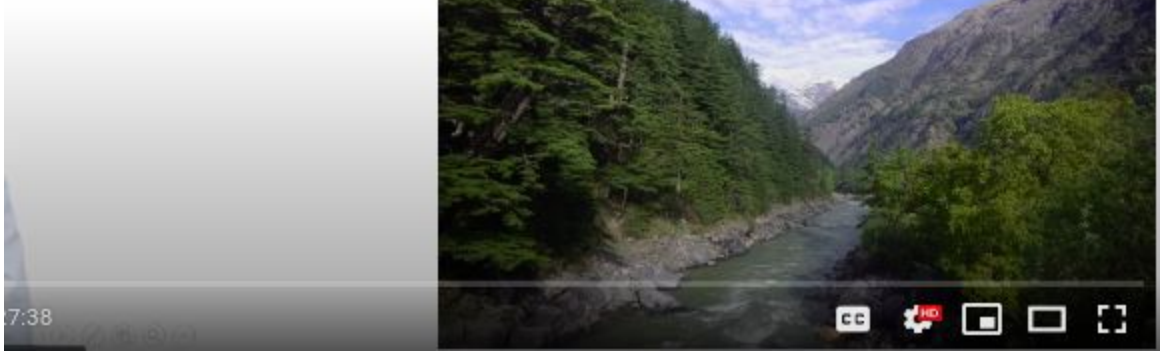
Vignetting

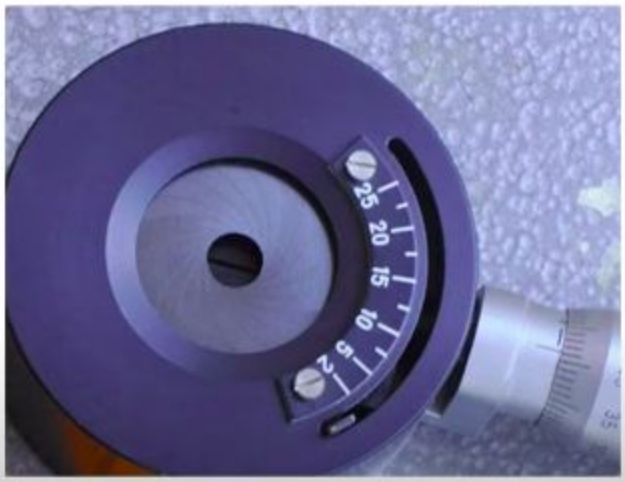


Amount of light reaching extremities of the object

Examples: Vignetting



Laboratory Stop



Size of aperture and f/number (f/#)

- How do we decide if an aperture is large or small?
  - Compare aperture size with the focal length called f/# for infinite conjugates (for object at infinity)

$$f/\# = f/D$$

  - For objects not at infinity, working f/# used

rcise: Find out what is numerical aperture of a system and how it rel d f/#

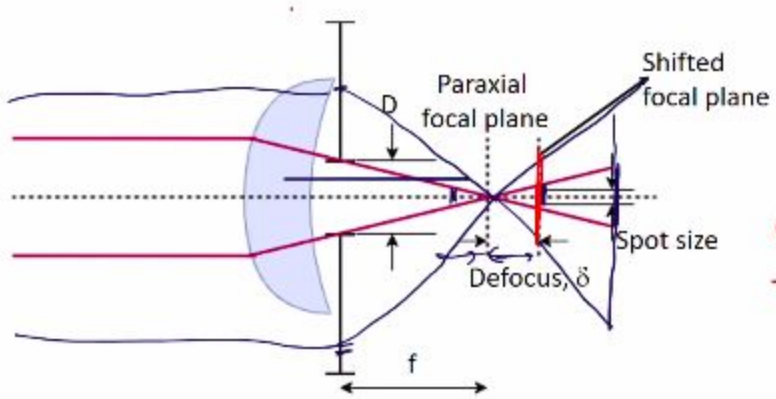
f# is also known as the speed of lens . Why?

f/D ($f_\#$): Speed of a lens

πr^2
$\pi (2r)^2$

- Higher f-number – smaller diameter
- Light increases 4x, when diameter doubles
  - f/2 has double the diameter of f/4
  - Amount of light- function of area
- Camera stops progress such that "1 stop" wider admits twice as much light (diameter increases by a factor of 1.414)
- Exposure time?

Aperture Stop and Depth of Focus



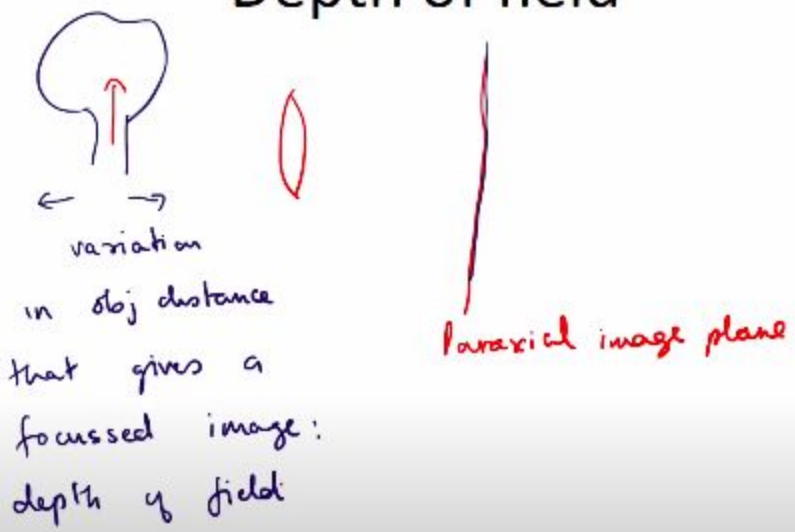
Pinhole camera – ∞ DO focus

- If image plane is not at the paraxial focal plane location then a point object is not sharply focused but produces a blur spot
- Size proportional to aperture stop (exit pupil) diameter
- Defocus for which spot size is acceptably small is the depth of focus
- Depth of focus shorter for larger aperture size

Spot size at focal plane = $\lambda f/D$

A pinhole camera has infinite depth of focus but has a large exposure time.

Depth of field



Aperture Stop and Depth of Field

f/2.8

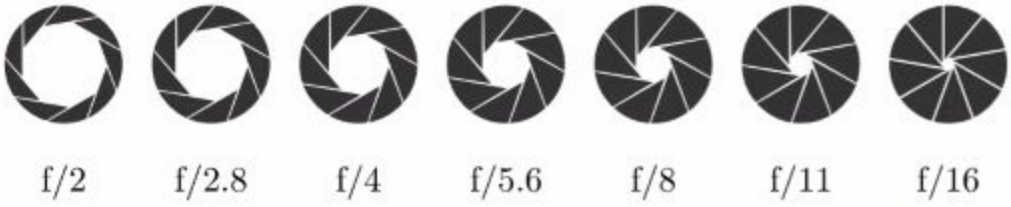
- At f/2.8, more light
- Shallow depth of field

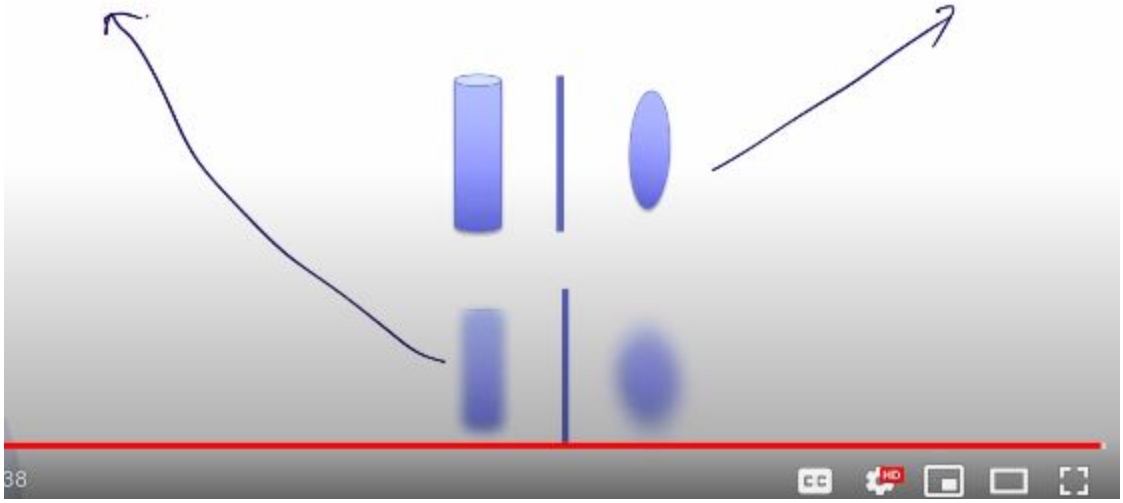
Aperture Stop and Depth of Field

- Greater depth of field
- F stops greater than f/2.8
- Shutter speed ten times longer

p://hyperphysics.phy-astr.gsu.edu/hbase/geoopt/stop.html

Depth of Field





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